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Quentin Rebergue - Graduate Software Engineer

Profile

Looking for a full-time position in software engineering with a focus on low-level systems / hardware integration.

MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded.

Fluent in French and English (TOEIC 915, C1). *Interests: watchmaking, music, hiking/running, cooking.*

Work Experience

- **Head of Teaching Assistants — YAKA / ACU Program, EPITA** 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses.

- **Web Developer, Various Clients** 09/2021 – 03/2024

  Designed, deployed, managed information system and apps for 10+ client missions.

- **Software Engineer Intern — Platform team, ARISTA** 02/2026 – 08/2026

  Implemented non-zero reset value management for ASIC Physical Table registers in the chip-agnostic infrastructure. Won 1st place at Arista's internal hackathon by building an AI agent optimization tool (-70% token usage).

School Projects

- **Distributed Task Queue** (Python, Redis, Docker, FastAPI) – Designed and implemented a distributed task queue system with Redis as the broker. Supports retries, dead-letter queues, and priority scheduling. Deployed with Docker Compose.
- **Tiger Compiler** (C++, LLVM, Flex, Bison) – Built compiler frontend in C++: lexing, parsing, binding, type checking, desugaring and LLVM IR.
- **RFID Driver** (C, Linux Kernel, SPI, QEMU) – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **My Docker** (C, Linux, Namespaces, cgroups, seccomp) – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **Unicycle CPU** (VHDL, FPGA, Intel Quartus) – Designed a 32-bit single-cycle ARM-like processor in VHDL (ALU, register file, instruction decoder, control unit, conditional branching), synthesized on a real FPGA with Intel Quartus, and validated with testbenches.

Education

Technical Expertise

- **Programming & Languages:** C, ASM, Shell, VHDL, CUDA, ADA, Rust, OOP.
- **Embedded & Hardware:** .
- **Other** / **DevOps:** Git, Perforce, Kubernetes, Security.
- **Systems:** Linux, Kernel, Drivers, Bootloader, Virtualization, Multithreading, GPU.
- **TypeScript:** TypeScript, JavaScript, Frontend, Backend, Node.js.